

Chakshu Gupta

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PROFESSIONAL SUMMARY

Computer Engineering student skilled in **C++, Python, DSA, OOP, DBMS, AI/ML, LLMs, and RAG**, with hands-on experience building backend and data-driven applications using **Python, SQL, APIs, Git, and modern AI frameworks**. Solved **400+ coding problems** on LeetCode.

EDUCATION

Thapar Institute of Engineering and Technology
Bachelor of Technology in Computer Engineering — CGPA: 8.40

Patiala, Punjab, India
Aug. 2023 – Present

WORK EXPERIENCE

Data Science Intern
Evoastra Ventures

Sept. 2025 – Nov. 2025
Remote

- Developed a machine learning pipeline for **telecom customer churn prediction**, implementing data ingestion, preprocessing, feature engineering, EDA, and model evaluation across **10K+ customer records**.
- Implemented and evaluated a **Gradient Boosting Classifier** achieving **89% accuracy**; improved high-risk customer identification by **25%** through hyperparameter tuning and **cross-validation**, applying structured problem-solving to improve model performance.

RESEARCH EXPERIENCE

GAN-based Watermark Removal | Research under Prof. Deep Maan

Aug. 2025 – Present

- Implemented **Pix2Pix**, **Residual GAN**, and **Spatial Pyramid Attention** based **GAN models** for watermark removal across varying **watermark types**, with a **research paper submitted for peer review**.
- Achieved **44.29 dB PSNR** outperforming **baseline models** and demonstrated strong **generalization** across **watermark sizes, opacity variations, and image resolutions** under diverse **real-world degraded image conditions**.

PROJECTS

MediCure – AI Healthcare Platform | FastAPI, PostgreSQL, LangGraph, LangChain ↗*

Feb. 2026 – March 2026

- Built and deployed an end-to-end **AI-powered web application** with **FastAPI** backend and **PostgreSQL** database, integrating patient monitoring, medical records, appointment scheduling, and automated clinical alerts.
- Engineered **multi-agent backend workflows** for symptom analysis, risk assessment, health checks, and proactive monitoring using **LangGraph**, with persistent state, and **SSE-based communication** for responsive workflows.

Codebase Intelligence Assistant | RAG, APIs, Vector Search, LLMs

July 2026 – Aug. 2026

- Built an automated **code intelligence and semantic search system** using **n8n**, **GitHub API**, **Gemini embeddings**, and **Qdrant**, indexing repository files for efficient code retrieval.
- Developed a **GitHub issue triage workflow** using vector similarity search and an **LLM** to analyze root causes and generate fixes; resolved **3072-dimensional vector** integration issues.

TrendyFire – Demand Forecasting System | Gen AI, Machine Learning, NLP, BI ↗*

Nov. 2025 – December 2025

- Built a **demand forecasting platform** integrating data collection, preprocessing, business analytics, and forecasting workflows.
- Automated **forecasting, reporting, and visualization** workflows through programmatic data processing, enabling faster analysis and iterative evaluation of demand trends.

TECHNICAL SKILLS

Programming Languages: C++, Python, SQL
Computer Science: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Algorithm Design, Problem Solving
AI & Machine Learning: Machine Learning, Deep Learning, Generative AI, LLMs, RAG, NLP, Transformers, Computer Vision, Feature Engineering, Model Evaluation
Libraries & Frameworks: FastAPI, LangChain, LangGraph, PyTorch, TensorFlow, Scikit-Learn, pandas, NumPy, OpenCV, Matplotlib, Streamlit
Tools & Platforms: n8n, Qdrant, PostgreSQL, GitHub, Power BI, Jupyter Notebook, Google Colab

CERTIFICATIONS

Generative AI: Introduction and Applications — IBM [Credential](#)
Generative AI: Foundation Models and Platforms — IBM [Credential](#)
NVIDIA: Fundamentals of Deep Learning — Whizlabs [Credential](#)